Quantum Mechanics from Orbital Structure: The Schrödinger and Dirac Equations from Unitarity, Representations, and Conservation

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Abstract

We derive the fundamental wave equations of quantum mechanics — the Schrödinger, Klein–Gordon, and Dirac equations — from the same three-step orbital pattern used to derive special relativity (Paper 1) and general relativity (Paper 2): *symmetry group → conservation → uniqueness.* In the quantum regime (β), the conservation principle becomes **unitarity** (probability conservation), the symmetry group is the **Bargmann group** (central extension of the Galilei group) for non-relativistic mechanics or the **Poincaré group** for relativistic mechanics, and the uniqueness theorem is **Schur's lemma** applied to the Casimir operators.

We prove three results: (i) the **unitarity–Noether correspondence** (Theorem 3.2): unitarity in quantum mechanics is the precise structural analog of Noether conservation in classical physics, connected by Stone's theorem; (ii) the Schrödinger equation is the *unique* evolution equation compatible with Bargmann equivariance, unitarity, and irreducibility, via the Casimir $C_2 = H - P^2/2m$ and Schur's lemma (Theorem 4.11); (iii) the **Bianchi–Casimir correspondence** (Theorem 6.1): the Bianchi identity $\nabla_\mu G^{\mu\nu} \equiv 0$ (which determines Einstein's equations in Paper 2) and the Casimir eigenvalue equation $C_i = \lambda_i \cdot \mathrm{Id}$ (which determines the wave equations) play structurally identical roles as *orbital integrability conditions.*

No Lagrangian is used. The theories are classified by a pair (κ, ρ): kernel dimension κ and coupling regime $\rho \in \{\alpha, \beta\}$. SR $= (10, \alpha)$; GR $= (<10, \alpha)$; QM $= (10, \beta)$; quantum gravity $= (<10, \beta)$.

Keywords: Schrödinger equation, Bargmann group, Wigner classification, unitarity, Noether conservation, orbital structure, Casimir operators, Stone–von Neumann, Lagrangian-free derivation

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1 Introduction

1.1 The standard derivations

The Schrödinger equation was introduced heuristically [1] via the de Broglie–Hamilton–Jacobi analogy: replace the classical energy E by $i\hbar\partial_t$ and momentum p by $-i\hbar\nabla$ in

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$E = p^2/2m + V$. The Dirac equation [2] was obtained by factorizing the Klein–Gordon operator. Feynman's path integral [3] derives the Schrödinger equation from a Lagrangian via the sum over paths. All three approaches involve either heuristic substitution rules or variational principles.

A different tradition derives wave equations from group representations. Wigner [4] classified elementary particles as irreducible unitary representations of the Poincaré group. Bargmann [5] identified the central extension of the Galilei group as the correct symmetry for non-relativistic quantum mechanics, with mass as the central charge. Lévy-Leblond [6] constructed Galilei-invariant wave equations for arbitrary spin. Musielak [7] derived infinite sets of wave equations from the extended Galilean group.

1.2 The question

Papers 1 and 2 [14, 15] derived SR and GR from the orbital pattern: symmetry → conservation → uniqueness, without variational principles. Does the *same* pattern produce quantum mechanics? Specifically:

- What plays the role of "conservation" in the quantum regime?
- What plays the role of "uniqueness"?
- Is the structural parallel with GR precise or merely analogical?

1.3 The answers

Conservation becomes **unitarity**: $\|U(t)\psi\| = \|\psi\|$ for all t (probability is conserved). Uniqueness becomes **Schur's lemma**: on an irreducible representation, the Casimir operators take definite scalar values, which determine the Hamiltonian and hence the wave equation.

The parallel with GR is precise, not analogical. The Bianchi identity (Paper 2) and the Casimir eigenvalue equation (this paper) play the same structural role: both are *orbital integrability conditions* that constrain the dynamics from the group structure alone.

2 The Regime Transition: Classical to Quantum

Definition 2.1 (Two regimes of equivariance)**.** Let G be a symmetry group acting on a state space $\mathcal{S}$.

- **Regime** (α) (classical): The state is a point $s \in \mathcal{S}$. Observables are functions $f : \mathcal{S} \to \mathbb{R}$. The group acts by diffeomorphisms: $g \cdot s \in \mathcal{S}$. The coupling between symmetry and observation is *deterministic*: a transformation g produces a definite change $f(g \cdot s)$.
- **Regime** (β) (quantum): The state is a ray $[\psi] \in \mathbb{P}\mathcal{H}$ in a projective Hilbert space. Observables are self-adjoint operators $\hat{A}$ on $\mathcal{H}$. The group acts by unitary operators: $g \mapsto U(g)$. The coupling is *stochastic*: a measurement of $\hat{A}$ on state ψ yields eigenvalue a_i with probability $|\langle a_i|\psi\rangle|^2$.

Remark 2.2 (What changes between regimes). The *group* G is the same (Poincaré or Galilei). What changes is the *representation space* and the *coupling type*:

	Regime (α)	**Regime** (β)
State	Point $x \in \mathcal{M}$	Vector $\psi \in \mathcal{H}$
Observable	Function $f(x)$	Operator $\hat{A}$
Symmetry action	Diffeomorphism ψ^*	Unitary $U(g)$
Measurement	Deterministic $f(x)$	Stochastic $\|\langle a_i\|\psi\rangle\|^2$

The structural constant of regime (β) is $\hbar$: it parameterizes the central extension (Bargmann) and sets the scale of non-commutativity. Like c (Paper 1) and G, Λ (Paper 2), $\hbar$ is an *input parameter*, not derived from the framework.

3 Unitarity as Quantum Conservation

Theorem 3.1 (Stone's theorem [8, 9]). *Every strongly continuous one-parameter unitary group $\{U(t)\}_{t\in\mathbb{R}}$ on a Hilbert space $\mathcal{H}$ has the form:*

$$U(t) = e^{-iHt/\hbar} \tag{1}$$

where H is a (possibly unbounded) self-adjoint operator on $\mathcal{H}$, called the generator*. Conversely, every self-adjoint operator generates a strongly continuous unitary group.*

Theorem 3.2 (Unitarity–Noether correspondence). *The structural analog between classical conservation and quantum unitarity is:*

Concept	***Classical (regime α)***	***Quantum (regime β)***
Structure preserved	*Metric g (by isometries)*	*Norm $\\|\psi\\|$ (by unitaries)*
Symmetry generator	*Killing vector ξ*	*Self-adjoint operator $\hat{A}$*
What it generates	*Isometric flow ϕ_t^ξ*	*Unitary group $e^{-i\hat{A}t/\hbar}$*
Conservation	*$dQ_\xi/dt = 0$ (charge const.)*	*$d\langle\hat{A}\rangle/dt = 0$ (expect. const.)*
Algebraic condition	*$\mathcal{L}_\xi g = 0$ (Killing eq.)*	*$[\hat{A}, H] = 0$ (commutator)*

The bridge is Stone's theorem: a self-adjoint generator $\hat{A}$ produces a unitary group (norm preservation), just as a Killing vector ξ produces an isometric flow (g-preservation). In both cases: a generator that commutes with the dynamics produces a conserved quantity.

Proof. **Classical side.** If ξ is Killing ($\mathcal{L}_\xi g = 0$) and $\nabla_\mu T^{\mu\nu} = 0$, then $Q_\xi = \int T^{\mu\nu}\xi_\nu \, \mathrm{d}\Sigma_\mu$ is constant in time (Paper 2, Proposition 2.4).

Quantum side. If $[\hat{A}, H] = 0$ (i.e., $\hat{A}$ commutes with the Hamiltonian), then $\frac{\mathrm{d}}{\mathrm{d}t}\langle\psi(t)|\hat{A}|\psi(t)\rangle = \frac{i}{\hbar}\langle\psi|[H, \hat{A}]|\psi\rangle = 0$. The expectation value is conserved.

The structural parallel: in both cases, the conservation law arises because the generator commutes with the dynamics ($\mathcal{L}_\xi g = 0$ classically; $[\hat{A}, H] = 0$ quantum-mechanically). Stone's theorem provides the precise dictionary: self-adjoint operator $\leftrightarrow$ unitary group $\leftrightarrow$ conserved expectation value. $\square$

Remark 3.3. We must be precise about the scope of this correspondence. Unitarity ($U^\dagger U = 1$, norm preservation) is the quantum analog of *symplecticity* (the classical flow preserves the symplectic form), not directly of Noether conservation (a specific charge is constant). The Noether analog in QM is the statement $[\hat{A}, H] = 0 \Rightarrow d\langle\hat{A}\rangle/dt = 0$, which is a *consequence* of unitarity, not the same thing. The correspondence is: unitarity provides the framework within which quantum Noether conservation operates, just as symplecticity provides the framework for classical Noether conservation. The two are *structurally analogous*, not identical.

4 The Non-Relativistic Case: Schrödinger from Bargmann

4.1 The Bargmann group

Definition 4.1 (Galilei group)**.** The Galilei group $\mathcal{G}_{\mathrm{Gal}}$ consists of transformations $(\tau, \vec{a}, \vec{v}, R)$ acting on spacetime by: $t \mapsto t+\tau$, $\vec{x} \mapsto R\vec{x}+\vec{v}t+\vec{a}$, where $\tau \in \mathbb{R}$ (time translation), $\vec{a} \in \mathbb{R}^3$ (spatial translation), $\vec{v} \in \mathbb{R}^3$ (boost), $R \in \mathrm{SO}(3)$ (rotation). Dimension: $1+3+3+3 = 10$.

Definition 4.2 (Bargmann group — central extension [5])**.** The Galilei group admits a non-trivial **central extension** by $\mathbb{R}$:

$$1 \to \mathbb{R} \to \widetilde{\mathcal{G}} \to \mathcal{G}_{\mathrm{Gal}} \to 1. \tag{2}$$

The extended (Bargmann) group $\widetilde{\mathcal{G}}$ has elements $(\theta, \tau, \vec{a}, \vec{v}, R)$ with composition:

$$(\theta_2, g_2) \cdot (\theta_1, g_1) = \left(\theta_2 + \theta_1 + \frac{m}{2\hbar}\omega(g_2, g_1),\ g_2 g_1\right) \tag{3}$$

where $\omega(g_2, g_1) = \vec{v}_2 \cdot R_2\vec{a}_1 - \vec{v}_2 \cdot \vec{v}_1\tau_1$ is Bargmann's 2-cocycle.

Theorem 4.3 (Bargmann algebra)**.** *The Lie algebra of $\widetilde{\mathcal{G}}$ has generators H (energy), P_i (momentum), K_i (boost), L_{ij} (angular momentum), and M (mass — central). The non-vanishing brackets include:*

$$[L_{ij}, P_k] = \delta_{ik}P_j - \delta_{jk}P_i, \tag{4}$$
$$[L_{ij}, K_k] = \delta_{ik}K_j - \delta_{jk}K_i, \tag{5}$$
$$[K_i, H] = P_i, \tag{6}$$
$$[K_i, P_j] = \delta_{ij}\, M, \tag{7}$$

where M is central: $[M, \cdot\,] = 0$. The bracket (7) is the defining relation of the central extension. It has no classical analog (for the unextended Galilei group, $[K_i, P_j] = 0$).

Remark 4.4 (Mass as central charge). In Paper 1, the speed of light c is the structural constant of the Poincaré kernel. Here, the mass m enters as the *central charge* of the Bargmann extension. In both cases, a fundamental physical quantity (speed, mass) is a *parameter of the symmetry algebra*, not a property of a particular object.

4.2 The three axioms for quantum mechanics

Axiom 4.5 (Bargmann equivariance)**.** Time evolution is implemented by a unitary representation $U : \widetilde{\mathcal{G}} \to \mathcal{U}(\mathcal{H})$ of the Bargmann group on a Hilbert space $\mathcal{H}$.

Axiom 4.6 (Unitarity)**.** The time evolution $U(t) = e^{-iHt/\hbar}$ preserves the norm: $\|U(t)\psi\| = \|\psi\|$ for all t and $\psi \in \mathcal{H}$. Equivalently, H is self-adjoint.

Axiom 4.7 (Irreducibility)**.** The representation U is irreducible: no proper closed subspace of $\mathcal{H}$ is invariant under all $U(g)$.

Remark 4.8. These axioms parallel the GR axioms of Paper 2: (Q1) $\leftrightarrow$ (A1) [group equivariance]; (Q2) $\leftrightarrow$ (A2) [conservation]; (Q3) $\leftrightarrow$ (A3) [a simplicity constraint selecting the minimal structure].

4.3 Uniqueness via Casimirs and Schur's lemma

Theorem 4.9 (Schur's lemma)**.** *If $\pi : G \to \mathcal{U}(\mathcal{H})$ is an irreducible unitary representation and $T : \mathcal{H} \to \mathcal{H}$ commutes with all $\pi(g)$ ($[T, \pi(g)] = 0$ for all $g \in G$), then $T = \lambda \cdot \mathrm{Id}$ for some scalar λ.*

Definition 4.10 (Casimir operators)**.** The **Casimir operators** of the Bargmann algebra are the elements of the center of the universal enveloping algebra $U(\widetilde{\mathfrak{g}})$. They commute with all generators and hence, on any representation, commute with all $U(g)$. For the Bargmann algebra, the Casimirs are:

$$C_1 = M, \tag{8}$$

$$C_2 = H - \frac{P^2}{2M}, \tag{9}$$

$$C_3 = \left(\vec{L} - \frac{\vec{K} \times \vec{P}}{M}\right)^2. \tag{10}$$

Here C_1 is the mass, C_2 is the internal (non-translational) energy, and C_3 encodes the spin.

Theorem 4.11 (Schrödinger equation from Bargmann representations)**.** *Axioms Q1–Q3 uniquely determine the free Schrödinger equation:*

$$\boxed{i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m}\nabla^2\psi + E_0\psi} \tag{11}$$

where E_0 is the internal energy (a representation-dependent constant). Adding an external potential $V(\vec{x})$ (which breaks Galilean symmetry locally) gives the full Schrödinger equation $i\hbar\partial_t\psi = (-\hbar^2\nabla^2/2m + V)\psi$.

Proof. The proof proceeds in four steps, mirroring the structure of the GR derivation (Paper 2).

Step 1 (Casimir eigenvalues from Schur). By Axiom Q3 (irreducibility), Schur's lemma (Theorem 4.9) applies to each Casimir:

$$C_1 = m \cdot \mathrm{Id}, \qquad C_2 = E_0 \cdot \mathrm{Id}, \qquad C_3 = s(s+1)\hbar^2 \cdot \mathrm{Id}, \tag{12}$$

where $m > 0$ is the mass, $E_0 \in \mathbb{R}$ is the internal energy, and $s \in \{0, \frac{1}{2}, 1, \ldots\}$ is the spin.

Step 2 (Hamiltonian from C_2). The Casimir equation $C_2 = E_0 \cdot \mathrm{Id}$ reads:

$$H - \frac{P^2}{2m} = E_0 \cdot \mathrm{Id}, \qquad \text{hence} \qquad H = \frac{P^2}{2m} + E_0. \tag{13}$$

The Hamiltonian is *determined* by the representation theory (the Casimir eigenvalue), not postulated.

Step 3 (Position representation). By Axiom Q2 (unitarity) and the canonical commutation relations $[\hat{x}_i, P_j] = i\hbar\delta_{ij}$ (which follow from the Bargmann algebra via exponentiation [5]): the Schrödinger representation is $P_i = -i\hbar\partial/\partial x_i$ on $\mathcal{H} = L^2(\mathbb{R}^3)$. The Hamiltonian becomes:

$$H = \frac{P^2}{2m} + E_0 = -\frac{\hbar^2}{2m}\nabla^2 + E_0. \tag{14}$$

Step 4 (Time evolution). By Stone's theorem (Theorem 3.1), the unitary evolution $U(t) = e^{-iHt/\hbar}$ gives $i\hbar\partial_t\psi = H\psi$, which is the Schrödinger equation (11). □

Remark 4.12 (What was not used). No action was minimized. No classical Hamiltonian was "quantized." The Schrödinger equation emerged from: (1) the Bargmann group (symmetry), (2) unitarity (conservation), (3) Schur's lemma (uniqueness). The Casimir $C_2 = H - P^2/2m$ plays the role of the Bianchi identity in GR: it is the orbital constraint that determines the dynamics.

We acknowledge that the derivation of the Schrödinger equation from the Bargmann group is a well-known result (Bargmann [5], Lévy-Leblond [6]). The novelty of this paper lies not in the derivation itself but in: (a) placing it in the three-step orbital pattern alongside GR (Paper 2); (b) the unitarity–Noether correspondence (Theorem 3.2); (c) the Bianchi–Casimir correspondence (Theorem 6.1).

Remark 4.13 (Stone–von Neumann uniqueness). The position representation used in Step 3 (i.e., $P_i = -i\hbar\partial/\partial x_i$ on $L^2(\mathbb{R}^3)$) is not merely one possible choice: the **Stone–von Neumann theorem** [19, 20] guarantees that it is the *unique* (up to unitary equivalence) irreducible representation of the canonical commutation relations $[\hat{x}_i, P_j] = i\hbar\delta_{ij}$ on a separable Hilbert space (in the Weyl exponentiated form). This provides a second uniqueness result, complementing Schur's lemma: Schur determines the Hamiltonian ($H = P^2/2m + E_0$); Stone–von Neumann determines the representation space ($L^2(\mathbb{R}^3)$). Together, they completely fix the Schrödinger equation.

Remark 4.14 (The Bargmann cocycle is the Galilei-boost phase). The non-triviality of the central extension (Definition 4.2) is directly observable. Under a Galilei boost $\vec{x}' = \vec{x} - \vec{v}\,t$, $t' = t$, the free Schrödinger equation $i\hbar\partial_t\psi = -\frac{\hbar^2}{2m}\nabla^2\psi$ is covariant only if the wavefunction transforms with a phase (in the standard convention)

$$\psi'(\vec{x}', t') = \exp\Big(\tfrac{i}{\hbar}\big(m\,\vec{v}\cdot\vec{x} - \tfrac{1}{2}mv^2 t\big)\Big)\,\psi(\vec{x}, t).$$

Direct substitution confirms covariance, and the phase cannot be gauged away. This phase *is* Bargmann's 2-cocycle $\frac{m}{2\hbar}\omega$ of Definition 4.2, with the mass m as central charge: composing two boosts, the phases combine with the cocycle discrepancy $\propto m\,\omega(g_2, g_1)$, which is precisely the obstruction to the Galilei group acting by a genuine (non-projective) representation. The unextended Galilei group ($[K_i, P_j] = 0$) carries no such phase and hence no nontrivial mass; the central extension is therefore not a technical convenience but the algebraic home of mass.

Remark 4.15 (The potential V). The external potential $V(\vec{x})$ breaks the translational symmetry of the Galilei group (just as the stress-energy tensor $T_{\mu\nu}$ breaks the Poincaré symmetry in GR). In both cases, the symmetry-breaking source is an input to the equation, not derived from the framework.

5 The Relativistic Case: Wigner Classification

For relativistic quantum mechanics, the Bargmann group is replaced by the Poincaré group, and the Casimirs change accordingly.

Theorem 5.1 (Poincaré Casimirs)**.** *The irreducible unitary representations of the Poincaré group are classified by two Casimir invariants:*

$$C_1 = P^\mu P_\mu = -m^2c^2, \tag{15}$$
$$C_2 = W^\mu W_\mu = -m^2c^2\, s(s+1)\hbar^2, \tag{16}$$

where $W^\mu = \frac{1}{2}\epsilon^{\mu\nu\rho\sigma} M_{\nu\rho} P_\sigma$ is the Pauli–Lubanski pseudovector. The label (m, s) specifies the mass and spin.

Theorem 5.2 (Wave equations from Wigner classification)**.** *On an irreducible representation with mass m and spin s, the Casimir eigenvalue $C_1 = -m^2c^2$ determines the wave equation. In the position representation ($P_\mu = -i\hbar\partial_\mu$):*

$$C_1\,\psi = -m^2c^2\,\psi \qquad \Longrightarrow \qquad \left(\Box - \frac{m^2c^2}{\hbar^2}\right)\psi = 0, \tag{17}$$

where $\Box = \eta^{\mu\nu}\partial_\mu\partial_\nu = -c^{-2}\partial_t^2 + \nabla^2$ is the d'Alembertian. This is the ***Klein–Gordon equation****.*

For different spins, the little group (stabilizer of a reference momentum) determines the field content:

m	s	***Little group***	***Wave equation***
>0	0	SO(3)	*Klein–Gordon:* $(\Box - m^2c^2/\hbar^2)\phi = 0$
>0	$\frac{1}{2}$	SU(2)	*Dirac:* $(i\hbar\gamma^\mu\partial_\mu - mc)\psi = 0$
$=0$	1	ISO(2)	*Maxwell:* $\partial_\mu F^{\mu\nu} = 0$
$=0$	2	ISO(2)	*Linearized Einstein*

Proof for Klein–Gordon. The quadratic Casimir on the representation: $C_1 = P^\mu P_\mu = (-i\hbar\partial^\mu)(-i\hbar\partial_\mu) = -\hbar^2\Box$. By Schur's lemma on the irreducible representation: $C_1 = -m^2c^2 \cdot \mathrm{Id}$. Hence $-\hbar^2\Box\psi = -m^2c^2\psi$, i.e. $\Box\psi = (m^2c^2/\hbar^2)\psi$, giving $(\Box - m^2c^2/\hbar^2)\psi = 0$. □

Proof for Dirac. The Dirac equation arises from factorizing the Klein–Gordon operator. Taking γ-matrices with the Clifford algebra $\{\gamma^\mu, \gamma^\nu\} = -2\eta^{\mu\nu}$ (so that $(i\hbar\gamma^\mu\partial_\mu)^2 = \hbar^2\Box$), one has $(i\hbar\gamma^\mu\partial_\mu + mc)(i\hbar\gamma^\nu\partial_\nu - mc) = \hbar^2\left(\Box - \frac{m^2c^2}{\hbar^2}\right)$. Hence the first-order equation $(i\hbar\gamma^\mu\partial_\mu - mc)\psi = 0$ — the Dirac equation — implies Klein–Gordon, and its solutions form a spin-$\frac{1}{2}$ representation of the Poincaré group [4, 6]. □

Remark 5.3 (Spin without relativity). Lévy-Leblond [6] showed that the same factorization procedure applied to the Bargmann Casimir $C_2 = H - P^2/2m$ produces a four-component equation (the Lévy-Leblond equation) whose solutions carry spin-$\frac{1}{2}$ and yield the correct gyromagnetic ratio $g = 2$. Spin is a consequence of the *rotation subgroup* SO(3) (common to both Galilei and Poincaré), not of relativity.

6 The Bianchi–Casimir Correspondence

This section contains the central structural result of the paper.

Theorem 6.1 (Bianchi–Casimir correspondence)**.** *The Bianchi identity (Paper 2) and the Casimir eigenvalue equation (this paper) play structurally identical roles in their respective theories:*

Role	***GR (Paper 2)***	***QM (this paper)***
Symmetry group	$\mathrm{Diff}(\mathcal{M})$	*Bargmann / Poincaré*
State space	*Lorentzian metrics* g	*Hilbert space* $\mathcal{H}$
Equivariance	*Natural concomitant*	*Unitary representation*
Conservation	$\nabla_\mu T^{\mu\nu} = 0$	$\|U\psi\| = \|\psi\|$
Orbital constraint	$\nabla_\mu G^{\mu\nu} \equiv 0$ ***(Bianchi)***	$C_i = \lambda_i \cdot \mathrm{Id}$ ***(Casimir)***
Uniqueness theorem	*Curiel/Navarro-Sancho*	*Schur's lemma*
Output	$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi T_{\mu\nu}$	$i\hbar\partial_t\psi = H\psi$

Both constraints are ***orbital integrability conditions****: they specify which orbit of the symmetry group the physical system occupies, and the uniqueness theorem converts this orbit specification into a unique equation.*

Proof. The proof establishes the structural parallel step by step.

Step 1 (Equivariance → constraint). In GR: Diff-equivariance (Axiom A1) requires $\mathcal{E}_{\mu\nu}$ to be a natural concomitant. The Bianchi identity $\nabla_\mu G^{\mu\nu} \equiv 0$ is a geometric identity constraining which natural concomitants are divergence-free. In QM: Bargmann equivariance (Axiom Q1) requires U to be a unitary representation. The Casimir operators C_i commute with all $U(g)$ by construction, constraining which representations are irreducible.

Step 2 (Constraint → uniqueness). In GR: the constraint (divergence-free + second-order + natural) determines $\mathcal{E} = aG + bg$ by Curiel's theorem. In QM: the constraint ($C_i = \lambda_i$ on an irreducible representation) determines $H = P^2/2m + E_0$ by Schur's lemma.

Step 3 (Uniqueness → equation). In GR: $\mathcal{E} = aG + bg = \kappa T$ gives the Einstein equation. In QM: $H = P^2/2m + E_0$ gives the Schrödinger equation via Stone's theorem.

Step 4 (Conservation is automatic). In GR: $\nabla_\mu(aG+bg)^{\mu\nu} = 0$ (Bianchi + metric compatibility) guarantees $\nabla_\mu T^{\mu\nu} = 0$. In QM: $H = H^\dagger$ (self-adjointness, forced by the Casimir being real) guarantees unitarity $U(t)^\dagger U(t) = 1$. □

Remark 6.2 (Depth of the correspondence). The parallel is not superficial. In both cases:

- (i) The constraint is an *identity* — it holds for all metrics (Bianchi) or for all irreducible representations (Casimir), not just specific solutions.
- (ii) The constraint reduces the space of possible equations from infinite-dimensional to finite-dimensional (the space of constants a, b for GR; the space of labels m, s, E_0 for QM).
- (iii) The constraint is *forced by the group structure*, not postulated. The Bianchi identity follows from the Jacobi identity for ∇; the Casimir eigenvalue follows from the definition of irreducibility.
- (iv) Conservation is a *consequence*, not a premise: it follows automatically from the equation determined by the constraint.

7 The Classification Grid

Theorem 7.1 (Two-parameter classification)**.** *The fundamental theories of physics are classified by the pair* (κ, ρ)*:*

	$\kappa = 10$	$\kappa < 10$
Regime (α)	***SR***	***GR***
Regime (β)	***QM***	***Quantum Gravity***

(18)

Each cell is determined by the three-step orbital pattern with the appropriate group, conservation principle, and uniqueness theorem. The transitions between cells are:

- *SR $\to$ GR: reduce κ from 10 to < 10 (introduce curvature).*
- *SR $\to$ QM: change regime $\alpha \to \beta$ (deterministic $\to$ stochastic).*
- *QM $\to$ QG: reduce κ (quantum mechanics on curved spacetime).*
- *GR $\to$ QG: change regime $\alpha \to \beta$ (quantize gravity).*

Remark 7.2 (Limits). QM $\to$ SR when $\hbar \to 0$ (the stochastic coupling $P(\cdot|g)$ concentrates on the deterministic value). QG $\to$ GR when $\hbar \to 0$. QG $\to$ QM when $\kappa \to 10$ (flat limit). The fourth corner (QG) is constrained by requiring consistency with all three known corners.

8 Discussion

8.1 What the framework does not do

(i) **$\hbar$ is not derived.** Like c (Paper 1) and G, Λ (Paper 2), $\hbar$ is a structural constant of the symmetry algebra (the central extension parameter). Its value is an empirical input.

(ii) **The measurement problem is not solved.** The framework provides a geometric language (measurement = projection onto orbits of the observable's symmetry group), but not a mechanism for wave function collapse. It is compatible with any interpretation (Copenhagen, many-worlds, etc.).

(iii) **Quantum field theory requires a separate treatment.** The multi-particle extension (Fock space, spin-statistics, CPT) requires the cluster decomposition principle, which goes beyond single-particle QM. This is addressed in Paper 3b [16].

8.2 The three-step pattern across the series

Step	Paper 1 (SR)	Paper 2 (GR)	Paper 3a (QM)
1. Group	Poincaré	Diff($\mathcal{M}$)	Bargmann / Poincaré
2. Conservation	$dQ/dt = 0$	$\nabla_\mu T^{\mu\nu} = 0$	$\|U\psi\| = \|\psi\|$
3. Uniqueness	Max-sym classif.	Curiel/N-S	Schur's lemma
Orbital constraint	Mass-shell	Bianchi	Casimir
Output	Lorentz + $E^2 = \dots$	$G + \Lambda g = 8\pi T$	$i\hbar\partial_t\psi = H\psi$
Lagrangian	Not used	Not used	Not used

8.3 Connection to Paper 1

The mass-shell $E^2 = p^2c^2 + m^2c^4$ (Paper 1, Theorem 5.3, from the Poincaré coadjoint orbit) and the Klein–Gordon equation $(\Box - m^2c^2/\hbar^2)\phi = 0$ (this paper, Theorem 5.2) arise from the *same* Casimir $C_1 = P^\mu P_\mu = -m^2c^2$. The difference: in regime (α), the Casimir gives an orbit constraint on a *point* (the 4-momentum lives on the mass hyperboloid); in regime (β), it gives an operator equation on a *wave function* (the field satisfies the Klein–Gordon equation). The group structure is identical; the representation space changes.

9 Conclusion

We have derived the Schrödinger, Klein–Gordon, and Dirac equations from three axioms (Bargmann/Poincaré equivariance, unitarity, irreducibility) without any variational principle, following the same three-step orbital pattern as Papers 1 (SR) and 2 (GR).

The central results are: (i) the unitarity–Noether correspondence (Theorem 3.2), showing that quantum unitarity is the structural analog of classical Noether conservation; (ii) the Bianchi–Casimir correspondence (Theorem 6.1), showing that the Bianchi identity (GR) and the Casimir eigenvalue (QM) play structurally identical roles as orbital integrability conditions; (iii) the (κ, ρ) classification grid (Theorem 7.1), placing SR, GR, QM, and quantum gravity in a unified framework.

The pattern is now established for three of the four corners of the grid. The fourth corner — quantum gravity — is the subject of Paper 4 [17], which will formalize the universal pattern as a metatheorem and derive constraints on the missing theory.

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